



Name: _____

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Learning Objectives

- Classify a system of linear equations as consistent or inconsistent
- Determine whether a consistent system is dependent or independent
- Solve linear systems both graphically and analytically
- Use slope-intercept form to predict the behavior of a system's graph

Solve each system or analyze its behavior, then classify it as consistent independent, consistent dependent, or inconsistent.

1. Verify whether the ordered pair (2, 1) is a solution of the system.

$$\begin{cases} x + y = 3 \\ 2x - y = 3 \end{cases}$$

Answer: _____

2. Convert each equation to slope-intercept form and identify the slope and y-intercept.

$$2x + 3y = 12$$

Answer: _____

3. Solve the system graphically and classify it.

$$\begin{cases} y = x + 1 \\ y = -x + 3 \end{cases}$$

Answer: _____

4. Solve the system by substitution and classify the solution.

$$\begin{cases} y = 2x - 1 \\ 3x + y = 9 \end{cases}$$

Answer: _____

5. Solve the system by elimination and classify the solution.

$$\begin{cases} 2x + 3y = 7 \\ 4x - 3y = 5 \end{cases}$$

Answer: _____

6. Without solving, use the slopes and y-intercepts to determine whether the system has one solution, no solution, or infinitely many solutions.

$$\begin{cases} y = 3x + 2 \\ y = 3x - 5 \end{cases}$$

Answer: _____



7. Determine the number of solutions for the system after converting to slope-intercept form.

$$\begin{cases} 2x - y = 4 \\ 4x - 2y = 8 \end{cases}$$

Answer: _____

8. Solve the system analytically and classify it.

$$\begin{cases} x + 2y = 6 \\ x + 2y = 10 \end{cases}$$

Answer: _____

9. Classify the system by comparing slopes and y-intercepts after converting to slope-intercept form.

$$\begin{cases} 3x + y = 5 \\ 6x + 2y = 10 \end{cases}$$

Answer: _____

10. Solve the system and classify it as consistent independent, consistent dependent, or inconsistent.

$$\begin{cases} 5x - 2y = 4 \\ 3x + y = 9 \end{cases}$$

Answer: _____





This worksheet covers identifying solutions of linear systems: verifying a given solution, solving graphically, solving analytically (substitution and elimination), converting general form to slope-intercept form, comparing slopes and y-intercepts to predict one solution, no solution, or infinitely many solutions, and classifying systems as consistent/inconsistent and dependent/independent.

Solutions

1. Verify whether the ordered pair (2, 1) is a solution of the system.

$$\begin{cases} x + y = 3 \\ 2x - y = 3 \end{cases}$$

- Apply the substitution method to the system $[x + y = 3]$ and $[2x - y = 3]$.
- Reduce to one variable and solve to get $x = 2$.
- Substitute back to get $y = 1$.
- The solution is (2, 1); it checks in both original equations.

Answer: (2, 1)

2. Convert each equation to slope-intercept form and identify the slope and y-intercept.

$$2x + 3y = 12$$

- Subtract $2x$ from both sides to get $3y$ equals negative $2x$ plus 12 .
- Divide every term by 3 to isolate y .
- The slope is negative two-thirds and the y-intercept is 4 .

Answer: $y = -\frac{2}{3}x + 4$, $m = -\frac{2}{3}$, $b = 4$

3. Solve the system graphically and classify it.

$$\begin{cases} y = x + 1 \\ y = -x + 3 \end{cases}$$

- Apply the substitution method to the system $[y = x + 1]$ and $[y = -x + 3]$.
- Reduce to one variable and solve to get $x = 1$.
- Substitute back to get $y = 2$.
- The solution is (1, 2); it checks in both original equations.

Answer: (1, 2)

4. Solve the system by substitution and classify the solution.

$$\begin{cases} y = 2x - 1 \\ 3x + y = 9 \end{cases}$$

- Apply the substitution method to the system $[y = 2x - 1]$ and $[3x + y = 9]$.
- Reduce to one variable and solve to get $x = 2$.
- Substitute back to get $y = 3$.
- The solution is (2, 3); it checks in both original equations.

Answer: (2, 3)



5. Solve the system by elimination and classify the solution.

$$\begin{cases} 2x + 3y = 7 \\ 4x - 3y = 5 \end{cases}$$

→ Apply the substitution method to the system $[2x + 3y = 7]$ and $[4x - 3y = 5]$.

→ Reduce to one variable and solve to get $x = 2$.

→ Substitute back to get $y = 1$.

→ The solution is $(2, 1)$; it checks in both original equations.

Answer: (2, 1)

6. Without solving, use the slopes and y-intercepts to determine whether the system has one solution, no solution, or infinitely many solutions.

$$\begin{cases} y = 3x + 2 \\ y = 3x - 5 \end{cases}$$

→ Eliminate a variable using substitution.

→ The equations reduce to a false statement, so the lines are parallel.

→ Therefore the system has no solution.

Answer: No solution

7. Determine the number of solutions for the system after converting to slope-intercept form.

$$\begin{cases} 2x - y = 4 \\ 4x - 2y = 8 \end{cases}$$

→ Rewrite the first equation as y equals $2x$ minus 4 .

→ Rewrite the second equation as y equals $2x$ minus 4 .

→ Both equations represent the same line, so every point on the line is a solution.

→ The system is consistent and dependent with infinitely many solutions.

Answer: Infinitely many solutions; consistent and dependent

8. Solve the system analytically and classify it.

$$\begin{cases} x + 2y = 6 \\ x + 2y = 10 \end{cases}$$

→ Eliminate a variable using substitution.

→ The equations reduce to a false statement, so the lines are parallel.

→ Therefore the system has no solution.

Answer: No solution

9. Classify the system by comparing slopes and y-intercepts after converting to slope-intercept form.

$$\begin{cases} 3x + y = 5 \\ 6x + 2y = 10 \end{cases}$$

→ Rewrite the first equation as y equals negative $3x$ plus 5 .

→ Rewrite the second equation as y equals negative $3x$ plus 5 .

→ Both lines have the same slope and same y-intercept, so they are identical.

→ The system is consistent and dependent with infinitely many solutions.

Answer: Consistent and dependent; infinitely many solutions



10. Solve the system and classify it as consistent independent, consistent dependent, or inconsistent.

$$\begin{cases} 5x - 2y = 4 \\ 3x + y = 9 \end{cases}$$

→ Apply the substitution method to the system $[5x - 2y = 4]$ and $[3x + y = 9]$.

→ Reduce to one variable and solve to get $x = 2$.

→ Substitute back to get $y = 3$.

→ The solution is $(2, 3)$; it checks in both original equations.

Answer: $(2, 3)$

